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1.INTRODUCTION

1.1 Project Overview:

OrderOnTheGo – On-Demand Food Ordering Platform is a full-stack web application designed to simplify and digitize the process of ordering food from restaurants through a centralized online system. Traditional food ordering methods often involve limited menu visibility, manual coordination, longer waiting times, and inefficient communication between customers and restaurants. This project aims to address these challenges by providing a modern digital platform where users can browse restaurant menus, select food items, and place orders conveniently.

Customers can explore available restaurants, add items to their cart, and track order status in real time. Restaurant owners can manage menus, monitor incoming orders, and update order status efficiently through a dedicated dashboard. Administrators oversee platform operations, approve restaurant registrations, and ensure secure system management.

The application follows a modern client-server architecture using React.js for the frontend interface, Node.js and Express.js for backend services, and MongoDB Atlas for data storage and management.

1.2 Purpose:

The primary purpose of the OrderOnTheGo application is to enhance convenience, efficiency, and accessibility in online food ordering through a centralized digital solution. The project focuses on reducing manual ordering processes, improving communication between customers and restaurants, and delivering a seamless user experience.

Key objectives of the system include:

- Provide an intuitive platform for customers to discover restaurants, browse menus, and place food orders online.
- Allow restaurant owners to efficiently manage menu items, track orders, and update delivery status.
- Enable administrators to maintain platform governance and ensure secure operations.
- Deliver order status updates, notifications, and smooth user interactions.
- Build a scalable and user-friendly food ordering system using modern MERN stack technologies.

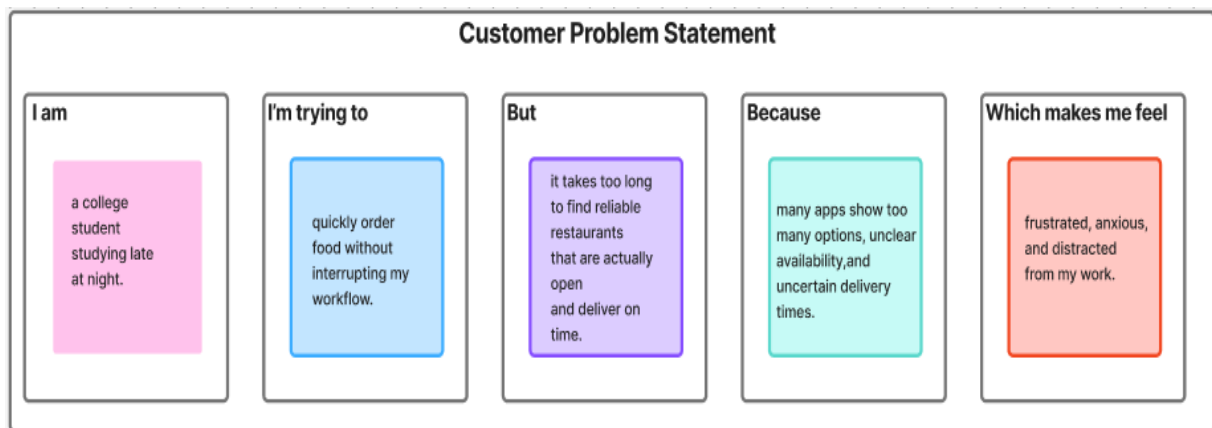
Overall, the project aims to bridge the gap between customers and restaurants by offering a reliable and structured food ordering solution that enhances user satisfaction and operational efficiency

2.IDEATION PHASE

2.1 Problem Statement:

In today's fast-paced lifestyle, ordering food from restaurants can often be time-consuming and inconvenient for both customers and restaurant owners. Traditional ordering methods may lack centralized menu visibility, real-time order tracking, and efficient communication between customers and restaurants. This can lead to delayed orders, incorrect requests, and reduced customer satisfaction. The OrderOnTheGo platform aims to address these challenges by providing a centralized digital food ordering solution

Problem Statement :

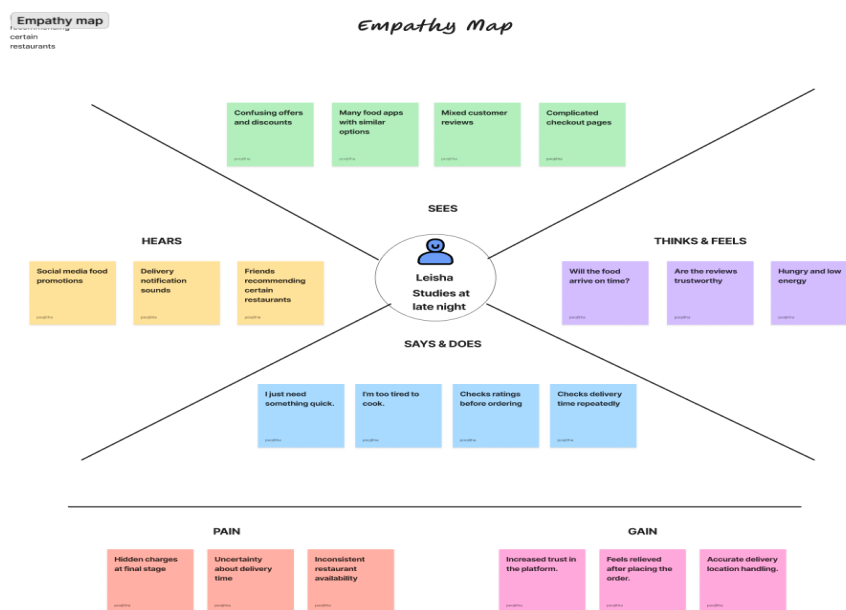


Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	a hungry user ordering food during unconventional hours.	make a confident decision about what to eat.	I cannot fully trust the reviews, pricing clarity, and final delivery cost shown upfront.	hidden charges and inconsistent experiences have reduced my trust in food delivery platforms.	uncertain and hesitant before placing my order.

PS-2	a time-sensitive user with low energy late at night.	complete my order in the fastest and simplest way possible.	the checkout process feels lengthy and requires multiple steps and repeated information.	the platform does not minimize effort or support quick decision-making.	mentally drained and impatient.
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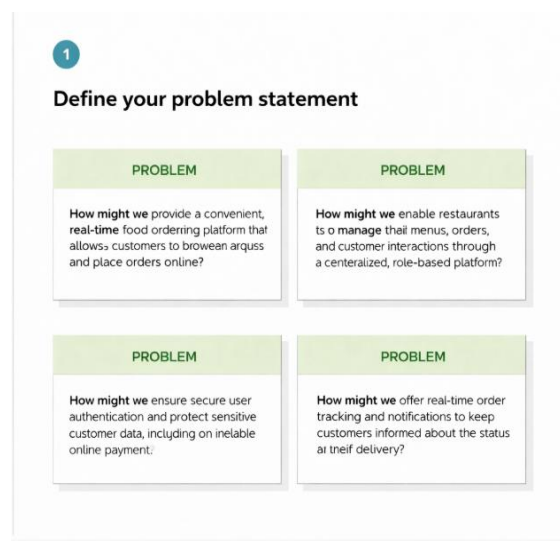
2.2 Empathy Map:

User: - Leisha(Studies At Late Night)



2.2 Brainstorm & Idea Prioritization:-

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

🕒 10 minutes

Poojitha Poshimshetti

Real-Time Convenience	Restaurant Management	Secure Payments	Secure Payments	Order Tracking
• Browse menus instantly • One-click "Place Order" button • Filter by cuisine, price, and ratings	• Restaurant dashboard for order management • Role-based login (Admin / Customer / Restaurant) • Easy menu updates by restaurant owners	• Secure online payment gateway • Sensitive data encryption • JWT authentication middleware	• Automated order status updates • Customer receives order notifications • Track delivery rider's live location	• Automated order status updates • Customer receives order notifications • Customer feedback after delivery

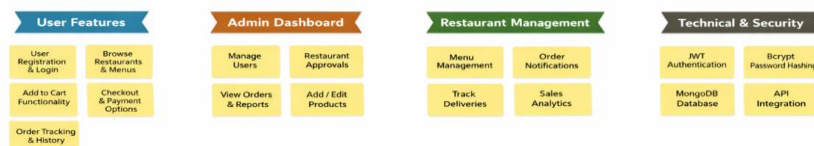
3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

🕒 20 minutes

TIP
Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mural.



Step-3: Idea Prioritization

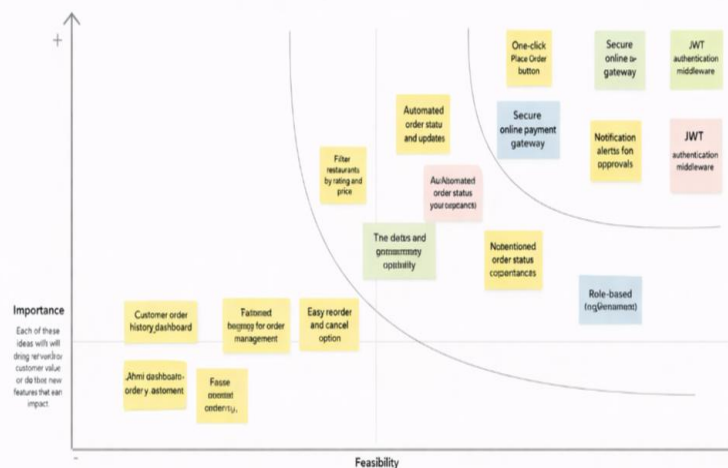
4 Step-4: Prioritize

Your team should all be on the same page about what's important moving forward. Place your idea you ran into this grid to determine which ideas are important and which are feasible.

🕒 20 minutes

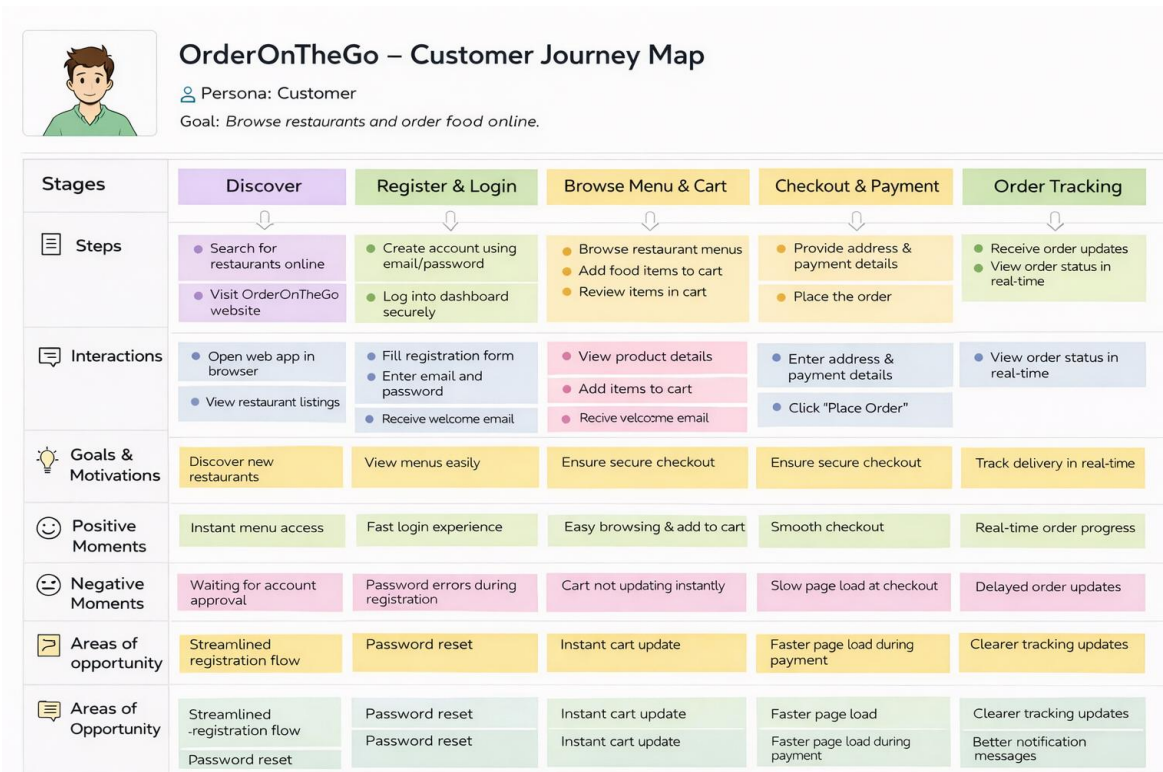
TIP

Prioritize ideas so you can see which ideas your team should focus on. Add tags to sticky notes to categorize ideas within the four quadrants to easily see key themes at a glance.



REQUIREMENT ANALYSIS

3.1 Customer Journey map:



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Users can register by providing name, email, password and Contact details. Store encrypted password in database.
FR-2	User Confirmation	Users can log in using registered credentials. System verifies credentials using JWT authentication Users can logout securely.
FR-3	Product Browsing & Filtering	Users can place orders from cart. Users enter delivery address and payment details. Order confirmation message is generated.
FR-4	Cart Management	Users can add products to cart.

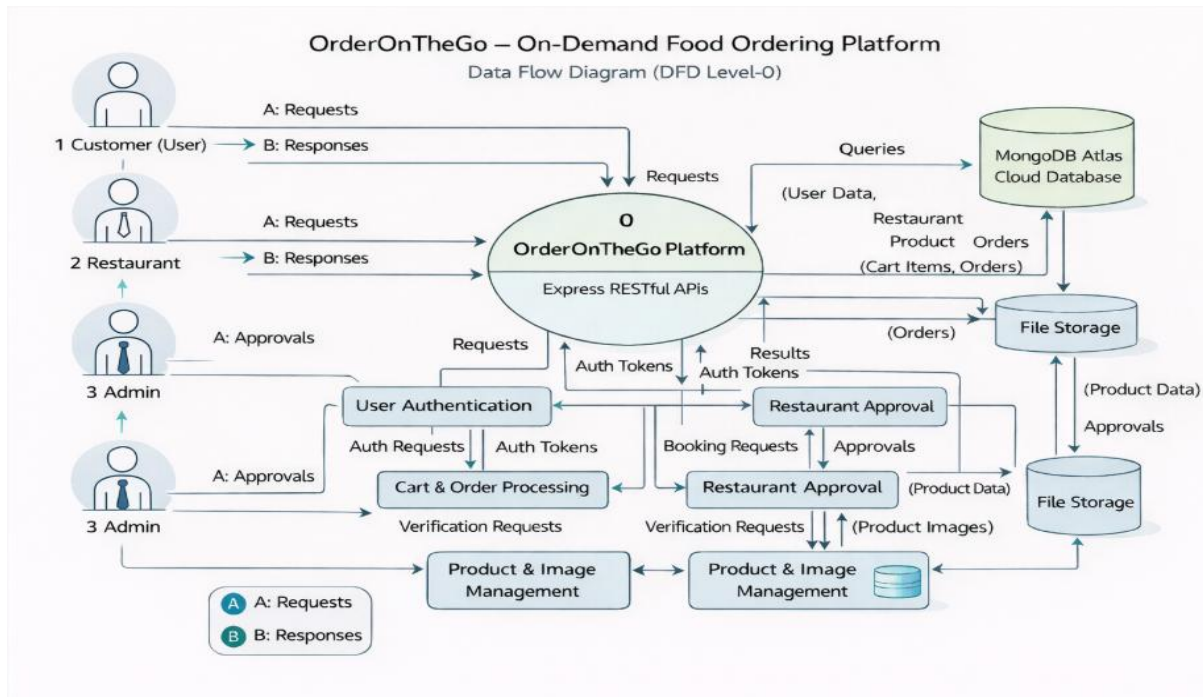
		Users can update quantity or remove items from cart
FR-5	Order Placement	Users can place orders from cart. Users enter delivery address and payment details. Order confirmation message is generated.
FR-6	Order Tracking & History	Users can view past orders in profile section. System stores order date, items, and payment status.
FR-7	Admin Management	Admin can login securely. Admin can manage users, products, restaurants, and orders.
FR-8	Restaurant Dashboard	Restaurants can add/edit/delete menu items. Restaurants can view received orders.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system provides a user-friendly interface with easy navigation, responsive design, and simple checkout process.
NFR-2	Security	User passwords are encrypted using hashing techniques. JWT authentication is implemented. Admin routes are protected.
NFR-3	Reliability	The system ensures accurate order processing and secure database transactions without data loss.
NFR-4	Performance	The application provides fast page loading, optimized API responses, and efficient database queries.
NFR-5	Availability	The system is available 24/7 and supports late-night food ordering.
NFR-6	Scalability	The system architecture supports adding more restaurants, users, and products without affecting performance.

3.3 Data Flow Diagram: -



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web User)	Registration	USN-1	As a customer, I can register using email and password	Account created & dashboard accessible	High	Sprint-1
Customer (Web User)	Authentication	USN-2	As a customer, I can log in securely	Successful login redirects to home page	High	Sprint-1
Customer (Web User)	Restaurant Listing	USN-3	As a customer, I can view available restaurants and food items	Restaurant list loads from database	High	Sprint-1
Customer (Web User)	Add to Cart	USN-4	As a customer, I can add food items to cart	Items appear in cart with quantity	High	Sprint-2
Customer (Web User)	Order Placement	USN-5	As a customer, I can place orders with address and payment method	Order status shows "Placed"	High	Sprint-2
Restaurant	Product Management	USN-6	As a restaurant, I can add/edit/delete food items	Products updated in dashboard	High	Sprint-3
Administrator	Restaurant Approval	USN-7	As an admin, I can approve restaurant registrations	Approved restaurant visible to users	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web User)	Notifications	USN-8	As a customer, I receive order updates	Order confirmation displayed on dashboard	Medium	Sprint-4
Customer (Web User)	Order History	USN-9	As a customer, I can view previous orders	Order history visible in profile	Medium	Sprint-5

3.4 Technology Stack: -

Technical Architecture: -

The OrderOnTheGo application follows a 3-Tier Client–Server Architecture designed to ensure scalability, maintainability, and secure communication between system components.

- **Presentation Layer (Frontend):** This layer handles menu browsing, cart management, order placement, dashboard views, and user profile management.
- **Application Layer (Backend):** The backend is built using Node.js and Express.js, which manage API requests, user authentication, order processing, restaurant management, and role-based access control. RESTful APIs enable communication between frontend and backend services.
- **Data Layer (Database):** The data layer uses MongoDB Atlas, a cloud-based NoSQL database.

The frontend communicates with backend REST APIs using Axios for efficient HTTP requests. Protected routes are secured using JWT authentication, ensuring that only authorized users can access role-specific dashboards and actions.

Table-1: Components & Technologies:

S. No	Component	Description	Technology
1.	User Interface	Web interface for users, restaurants, and admin dashboards	React.js, HTML, CSS, Bootstrap.
2.	Application Logic-1	Authentication & Role Management	Node.js, Express.js, JWT

3.	Application Logic-2	Food ordering, cart management, checkout process	Express.js REST APIs
4.	Application Logic-3	Order status tracking & notifications	Node.js Controllers
5.	Database	Stores users, restaurants, products, cart, and orders	MongoDB, Mongoose
6.	Cloud Database	Cloud-based NoSQL storage for scalable data access	MongoDB Atlas
7.	File Storage	Stores food images and restaurant media files	Local File System / Cloud Storage
8.	External API-1	Frontend to backend communication	Axios
9.	External API-2	Not Applicable	—

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build UI & Backend APIs	React.js, Express.js, Node.js
2.	Security Implementations	Role-based authentication, encrypted passwords, protected routes	JWT, bcryptjs, Middleware
3.	Scalable Architecture	Separation of frontend, backend, and database layers	REST Architecture, MongoDB
4.	Availability	Web application accessible anytime through browser	Node.js Server
5.	Performance	Fast API responses and asynchronous communication	Axios, Express.js

4. PROJECT DESIGN

4.1 Problem Solution Fit:



4.2 Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Customers often face difficulty ordering food quickly due to long waiting times, lack of centralized restaurant menus, and inefficient manual ordering methods. Restaurants struggle to manage orders and update menus efficiently, while admins need a secure system to monitor restaurant approvals, products, and user activities in one platform.
2.	Idea / Solution description	OrderOnTheGo is a MERN-stack web application that enables users to browse restaurants, add food items to cart, and place orders through a centralized online system. It provides role-based dashboards for Admin, Restaurant, and Customer with secure JWT authentication, product management, real-time order processing, and REST APIs connected to MongoDB Atlas.
3.	Novelty / Uniqueness	Role-based food ordering workflow, admin-controlled restaurant approvals, centralized product catalog, dynamic cart system,

		secure authentication, and a responsive user-friendly ordering interface that simplifies the food ordering experience.
4.	Social Impact / Customer Satisfaction	Reduces waiting time for customers, improves restaurant order management, provides clear menu visibility, and enhances user convenience through a fast and secure digital food ordering experience.
5.	Business Model (Revenue Model)	Commission-based model from restaurant orders, premium restaurant listings, promotional banners, and potential future integration of delivery tracking and subscription-based restaurant promotions.
6.	Scalability of the Solution	Built using a 3-tier MERN architecture (React frontend, Node.js/Express backend, MongoDB Atlas database), enabling scalable performance, easy deployment, and seamless expansion to support more users, restaurants, and features.

4.3 Solution Architecture:

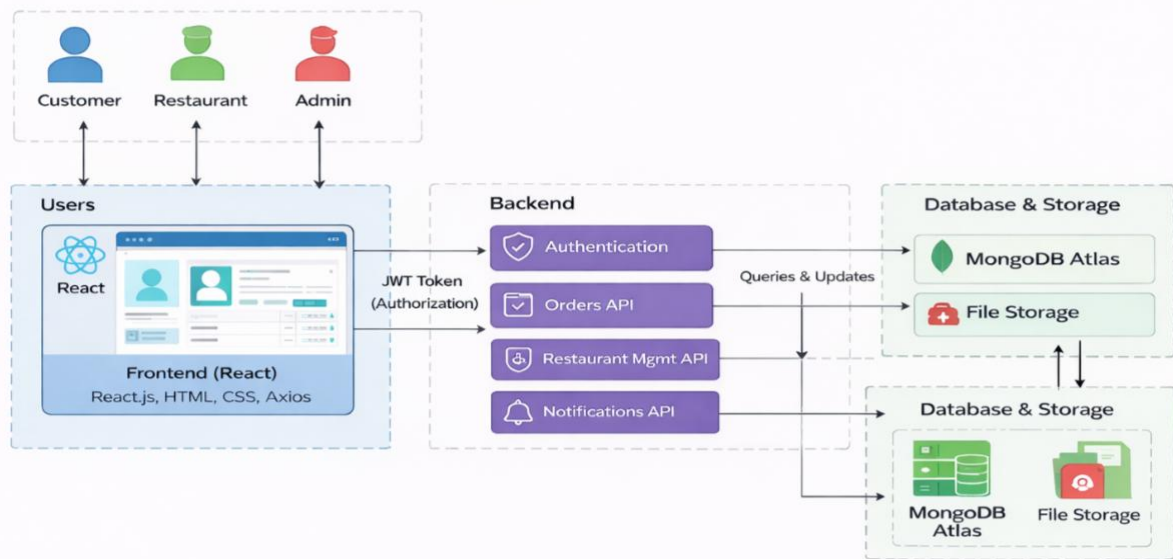


Figure-1: Architecture and data flow of the OrderOnTheGo MERN-stack food ordering platform.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning:

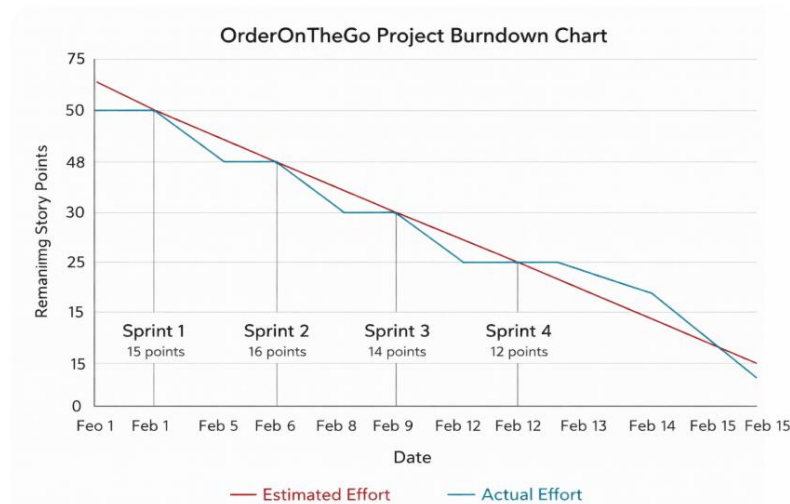
Product Backlog, Sprint Schedule, and Estimation :

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a customer, I can register using email and password to access the platform.	2	High	Team
Sprint-1	Authentication	USN-2	As a user, I can log in securely using JWT authentication.	2	High	Team
Sprint-1	Dashboard	USN-3	As a customer, I can view restaurant listings and food items on the dashboard.	3	High	Team
Sprint-2	Product Browsing	USN-4	As a customer, I can browse menus and view product details such as price and description.	5	High	Team
Sprint-2	Cart Management	USN-5	As a customer, I can add or remove food items from the cart before ordering.	3	Medium	Team
Sprint-3	Order Processing	USN-6	As a customer, I can place an order by providing address and payment details.	5	High	Team
Sprint-3	Restaurant Panel	USN-7	As a restaurant owner, I can add, update, and manage food products.	4	High	Team
Sprint-4	Admin Panel	USN-8	As an admin, I can approve restaurants and manage platform data.	3	Medium	Team
Sprint-4	Order History	USN-9	As a customer, I can view my order history and order status updates.	3	Medium	Team

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	15	4 Days	01 Feb 2025	04 Feb 2025	15	04 Feb 2025
Sprint-2	16	4 Days	05 Feb 2025	08 Feb 2025	16	08 Feb 2025

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-3	14	4 Days	09 Feb 2025	12 Feb 2025	14	12 Feb 2025
Sprint-4	12	3 Days	13 Feb 2025	15 Feb 2025	12	15 Feb 2025



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

Project Overview:

Project Name: DocSpot: OrderOnTheGo: On-Demand Food Ordering Platform

Project Description: OrderOnTheGo is a MERN-stack web application that enables customers to browse restaurants, add food items to cart, and place orders online with real-time menu visibility. The platform provides role-based dashboards for Admin, Restaurant, and Customer, secure JWT authentication, product management, and order status tracking using MongoDB Atlas.

Project Version: v1.0

Testing Period: 12 Feb 2025 to 15 Feb 2025

Testing Scope:

Features Tested:

- User Registration & Login
- Role-based Dashboard(Customer, Restaurant, Admin)
- Restaurant Listing & Product Menu

- Cart Management & Order Placement
- Restaurant Approval Workflow
- Order Status Updates & Notifications

User Stories Tested:

- USN-1 Registration
- USN-2 Login
- USN-4 Product Browsing & Cart
- USN-6 Order Placement
- USN-8 Notifications

Testing Environment:

URL/Location: <http://localhost:3000>

Credentials (if required): User → testuser@gmail.com / *****

Restaurant → retest@gmail.com / *****

Admin → admin@gmail.com / *****

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User Registration	Step1: Open Register Page Step2: Enter user details Step3: Submit	User account created successfully	Account created	Pass
TC-002	User Login	Step1: Enter email & password Step2: Click Login	Redirect to dashboard	Dashboard loaded	Pass
TC-003	View Restaurants	Step1: Login Step2: Open Home Dashboard	Restaurant list displayed	Restaurants visible	Pass
TC-004	Add to Cart	Step1: Select food item Step2: Click Add to Cart	Item added to cart	Cart updated	Pass
TC-005	Place Order	Step1: Open Cart Step2: Enter address & payment Step3: Submit Order	Order created successfully	Order pending	Pass
TC-006	Restaurant Product Management	Step1: Restaurant logs in Step2: Add new product	Product added to menu	Product visible	Pass
TC-007	Admin Approval	Step1: Admin logs in Step2: Approve restaurant	Restaurant status updated	Approved successfully	Pass
TC-008	Order Status Update	Step1: Restaurant updates order status	Status changes to Preparing/Delivered	Status updated	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Minor UI alignment issue in restaurant dashboard	Step-1: Login as Restaurant Step-2: Open Product Management page	Low	Closed	Fixed using CSS update
BG-002	Cart count not updating instantly after adding item	Step-1: Login as Customer Step-2: Add product to cart	Medium	Closed	Fixed using state refresh & API call optimization
BG-003	Slow image loading for food items on low internet	Open Home page with multiple products	Medium	In Progress	Image compression & lazy loading suggested

Sign-off:

Tester Name: Poshimshetti Poojitha

Date: 15 February 2025

Signature: Poshimshetti Poojitha

7. RESULTS

7.1 Output Screenshots:

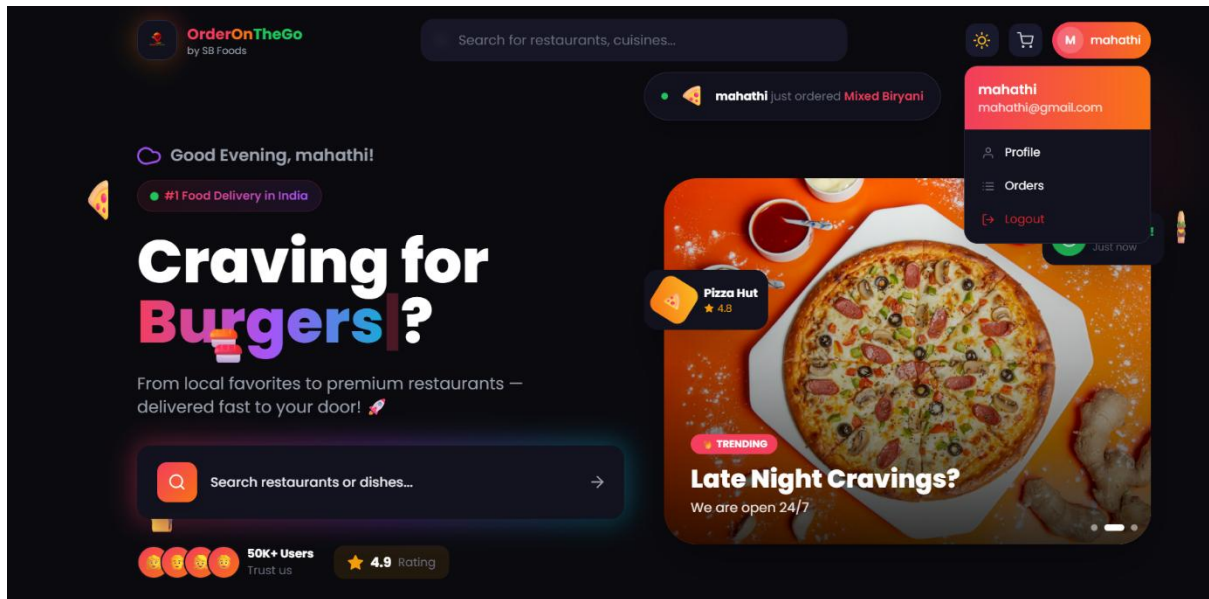


Figure-2: Customer Dashboard

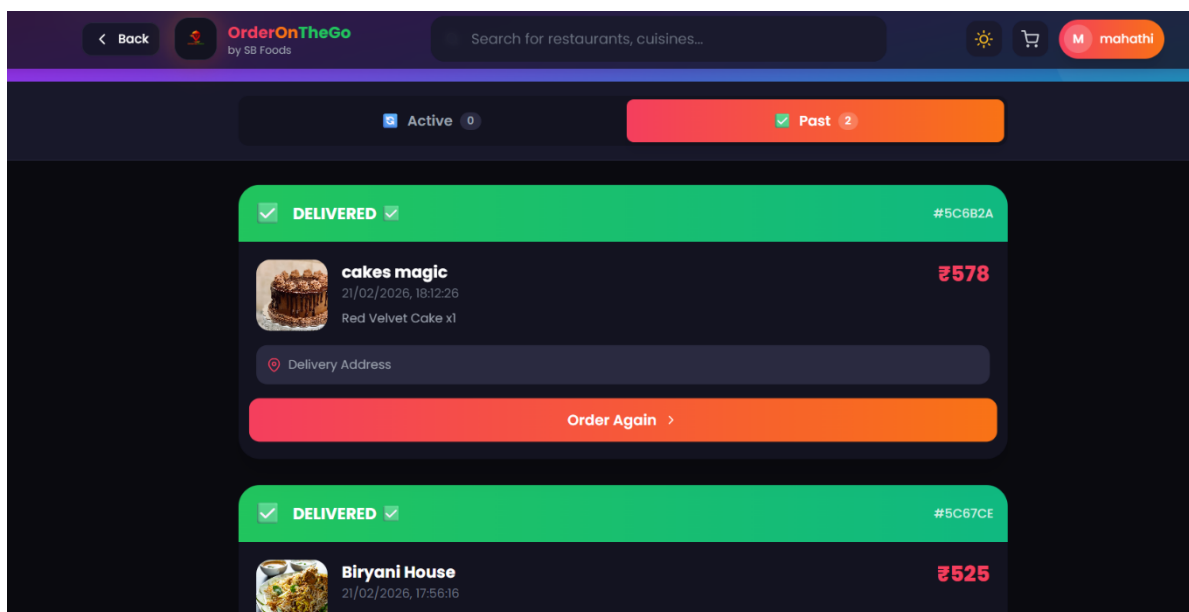


Figure-3: Order History

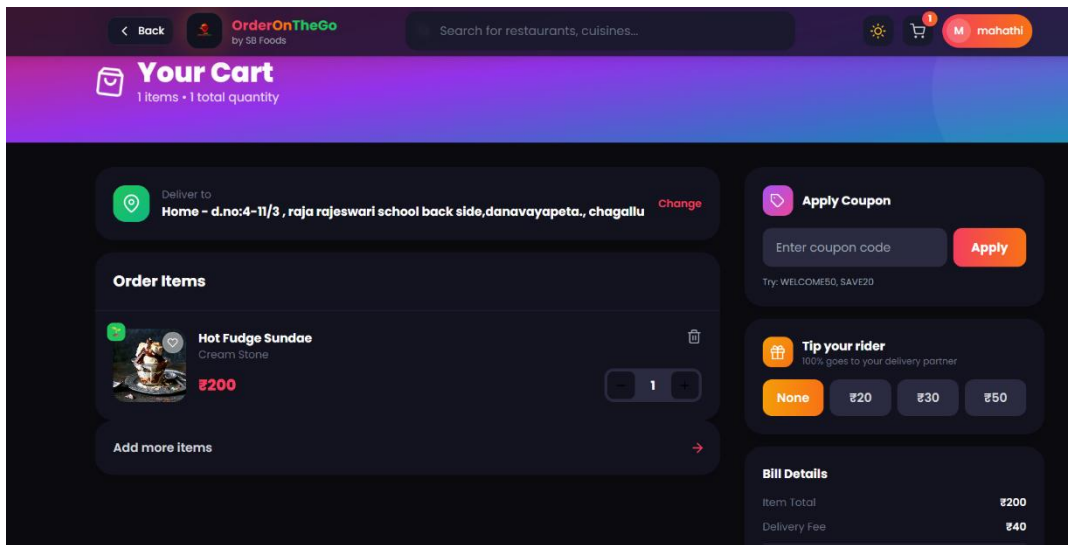


Figure-4: Cart Page

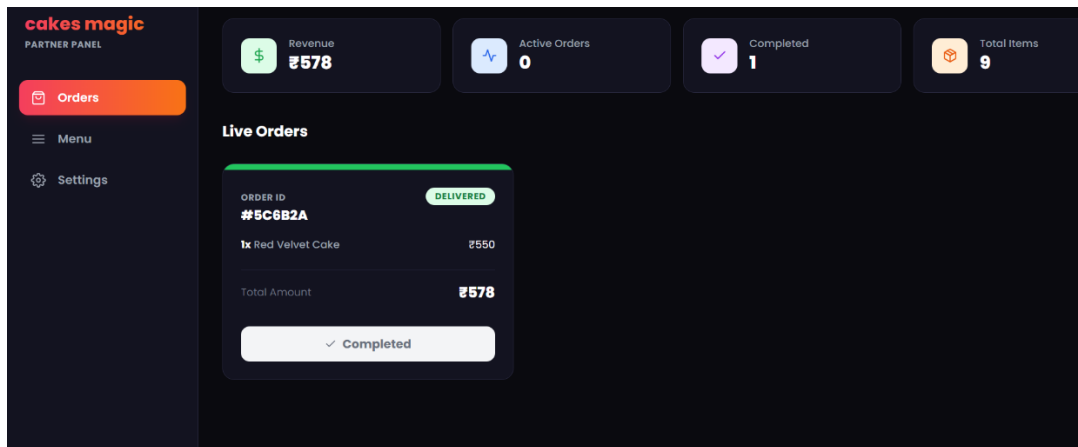


Figure-5: Restaurant Dashboard

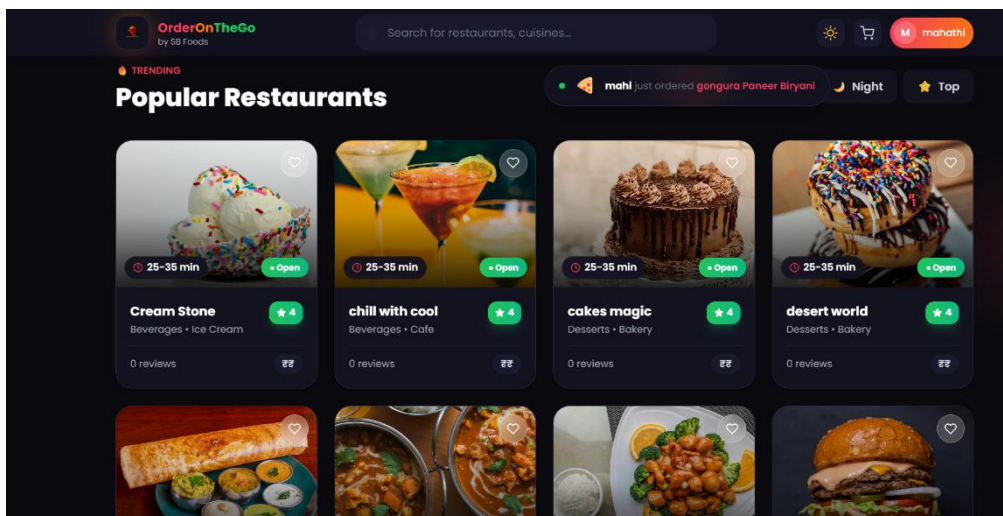


Figure-6: Popular Restaurants

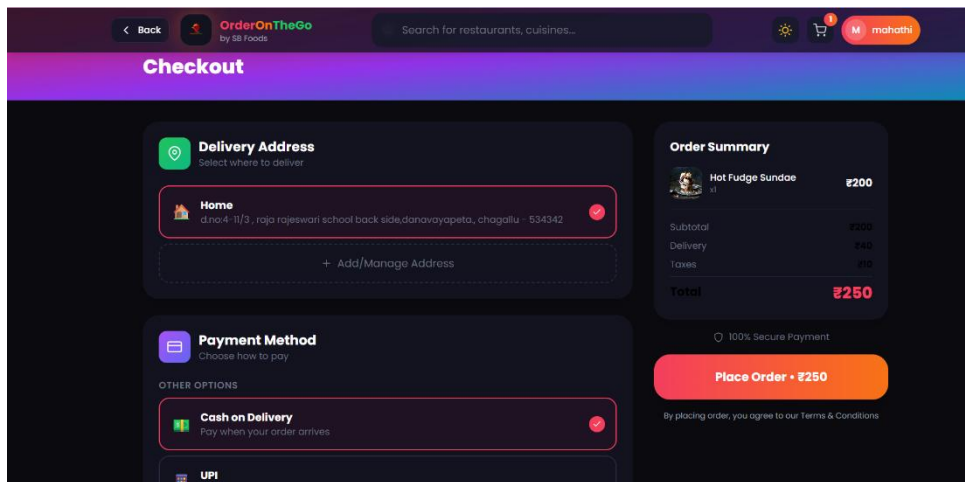


Figure-7: Checkout Page

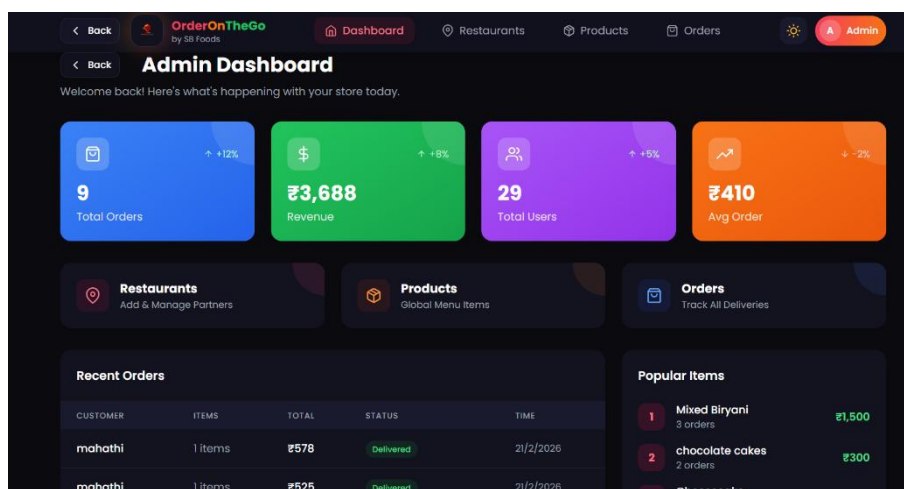


Figure-8: Admin Dashboard

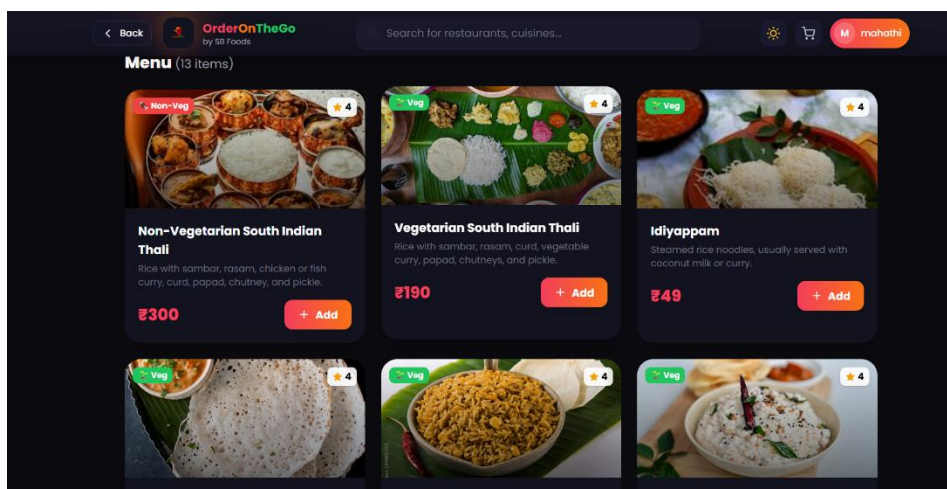


Figure-8: Restaurant Menu Listing Page

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Instant Food Ordering:** Customers can browse restaurant menus and place orders quickly without manual coordination, reducing waiting time.
- **Role-Based Access Control:** Separate dashboards for Admin, Restaurant Owner, and Customer improve workflow management and ensure organized system operations.
- **Cloud Database Integration:** Using **MongoDB Atlas** provides scalability, secure data storage, and remote accessibility for users and restaurants.
- **Secure Authentication:** JWT-based login protects user accounts and ensures that only authorized users can access protected features such as order management and admin controls.
- **Efficient Menu and Order Management:** Restaurant owners can manage food items, update menus, and track incoming orders through a centralized dashboard.
- **User-Friendly Interface:** Developed with React.js and modern UI frameworks, providing a responsive and intuitive ordering experience.

Disadvantages

- Requires a stable internet connection to browse menus and place food orders.
- Food image loading performance may vary depending on network speed and file size.
- Currently supports only a web-based platform (no dedicated mobile application).
- Real-time order notifications and live tracking features may require further enhancement for better user experience.

9. CONCLUSION

The OrderOnTheGo Food Ordering Platform successfully addresses the limitations of traditional food ordering methods by providing a centralized and user-friendly digital solution. By leveraging modern web technologies such as React.js, Node.js, Express.js, and MongoDB Atlas, the system enables seamless interaction between customers, restaurant owners, and administrators.

The platform enhances convenience by allowing users to browse menus, place orders efficiently, and track order status through an integrated dashboard. Restaurant owners can

manage food items and update order progress, while administrators maintain overall platform governance and security.

10. FUTURE SCOPE

- Mobile application development using React Native to enable food ordering on Android and iOS devices.
- Real-time push notifications using WebSockets or Firebase for instant order status updates.
- Payment gateway integration for secure online transactions using UPI, cards, or digital wallets.
- AI-based food recommendation system to suggest dishes and restaurants based on user preferences.
- Advanced analytics dashboard for restaurants and administrators to monitor orders and sales performance.

10. APPENDIX

GitHub Repository:

<https://github.com/poojitha-poshimshetti/OrderOnTheGo>

Live Demo / Project Video:

<https://drive.google.com/file/d/1afy-lmsBbqmpWzGKKzhpBWvjxShjguyc/view?usp=drivesdk>